

Literature Survey on Advanced PID Control and Optimization Techniques

Introduction

□ Ziegler-Nichols (ZN): Traditionally influential, the ZN step response method is based on simple process features.[\[1\]](#) However, it often lacks robustness and uses insufficient process information. Recent research has revisited ZN from the perspective of robust loop shaping to provide tuning rules that offer better performance for monotone step responses.[\[1\]](#)

□ Internal Model Control (IMC): This established approach designs PID controllers based on a process model.

- In coupled tank systems, IMC-based PID controllers have demonstrated superior performance and robustness compared to Z-N and Cohen-Coon methods by using a single tuning parameter—the closed-loop time constant[\[1\]](#).
- For DC-DC boost converters, a novel indirect IMC strategy has been introduced that eliminates the need for an IMC filter by focusing on a "plant shifting parameter," simplifying the tuning process for these non-minimum phase plants.[\[1\]](#)

Glossary (Keywords List with Meaning)

- **PID (Proportional-Integral-Derivative):** A control loop mechanism that calculates an error value as the difference between a desired setpoint and a measured process variable.
- **Ziegler-Nichols (ZN):** A classical heuristic method used to tune PID controllers based on step response or ultimate gain.
- **IMC (Internal Model Control):** A control strategy where the process model is explicitly embedded in the controller to simplify tuning to a single parameter.

- **Metaheuristic Optimization:** Nature-inspired algorithms (like GA or PSO) used to find optimal solutions in complex search spaces where traditional math may fail.
- **Pareto Optimality:** A state where no individual objective (e.g., tracking accuracy) can be improved without making at least one other objective (e.g., energy consumption) worse.
- **FOPID (Fractional Order PID):** An advanced PID controller using non-integer orders for integration and differentiation, offering more flexibility in tuning.

Search Methodology

The literature survey was conducted using IEEE Xplore, ScienceDirect, Springer, and Google Scholar databases. Keywords used include:

- PID Controller Optimization
- Internal Model Control PID
- Metaheuristic PID Tuning
- Fuzzy PID Controller
- NSGA-II Optimization
- Fractional Order PID

Inclusion Criteria:

- Peer-reviewed journals and conference papers
- Research published between 2000–2025
- Studies involving industrial and mechatronic control systems

Exclusion Criteria:

- Non-technical articles
- Studies without experimental or simulation validation
- Duplicate or outdated tuning approaches

The selected papers were analyzed based on robustness, settling time, overshoot, and optimization efficiency.

3. Science Effects

- **The Quarter Amplitude Damping Effect:** A tuning goal where each successive oscillation's amplitude is one-fourth of the previous one, used to stabilize industrial furnaces.
- **Right-Half-Plane (RHP) Zero Effect:** A phenomenon in boost converters that causes inherent instability and oscillations, requiring "plant shifting" or indirect IMC to stabilize.
- **Coupling Effect:** Observed in MIMO systems like the Quanser Aero 2, where the movement in one axis (pitch) affects the other (yaw).

4. Concepts, Rules, Principles, Laws

- **Robust Loop Shaping:** A principle used to modernize ZN tuning by integrating sensitivity constraints to ensure stability against process variations.
- **Fuzzy-Mamdani Logic:** A rule-based linguistic control principle that allows a controller to "think" like a human operator to adjust gains in real-time.
- **Natural Selection (GA):** The biological principle of "survival of the fittest" applied to evolve the most effective PID gains over multiple generations.
- **Social Behavior (PSO):** The principle of collective intelligence, mimicking bird flocking to converge on a global optimum gain.

5. Constants and Variables

- **K_p , K_i , K_d :** Proportional, Integral, and Derivative gains (variables adjusted during tuning).
- **λ (Closed-loop time constant):** The primary tuning variable in the IMC framework.
- **M_s (Sensitivity Peak):** A constant used as a constraint to ensure robustness in loop shaping.

- **$e(t)$ (Error):** The difference between the setpoint $r(t)$ and output $y(t)$.

6. Equations / Relationships

The fundamental PID control law in the time domain is defined as:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Performance Indices (Relationships):

- **ISE (Integral Squared Error):** $\int e^2(t) dt$ (Penalizes large errors).
- **ITAE (Integral of Time-weighted Absolute Error):** $\int t|e(t)| dt$ (Assesses long-term stability).

7. System of Equations / Model

For the **Internal Model Control (IMC)** framework, the controller $Q(s)$ is designed based on the plant model $G(s)$. In the case of **DC-DC Boost Converters**, the model must account for the non-minimum phase nature caused by the RHP zero to prevent instability.

8. What-If (Analysis)

- **What if the system is highly non-linear?** Traditional mathematical tuning (ZN) often fails; metaheuristic algorithms like Jellyfish Search (JSO) or Grey Wolf Optimizer (GWO) are required to find the global optimum.
- **What if we prioritize battery life over precision?** In MIMO systems like the Quanser Aero 2, engineers use the **Pareto Front** (via NSGA-II) to select an operating point that reduces control effort at the cost of some tracking accuracy.

Method	Application
IMC-PID	Coupled Tank Systems & Boost Converters
NSGA-II	Aerospace (Quanser Aero 2)
JSO & HHO	Paper Machines & Frequency Shifted IMC
ACO-GA (Hybrid)	Quarter-Car Active Suspension
Fuzzy-Mamdani	HVAC / Air Conditioners

Research Gaps Identified

Although advanced PID tuning methods improve system performance, several challenges still exist:

- Classical PID tuning methods lack adaptability for highly nonlinear systems.
- Metaheuristic optimization techniques often require high computational time.
- Many studies optimize only one parameter, such as overshoot or settling time, rather than multi-objective performance.
- Real-time implementation of AI-based controllers remains difficult in embedded industrial systems.
- Existing studies provide limited integration between intelligent optimization and energy-efficient control.

These gaps motivate the development of a robust and intelligent PID optimization framework capable of achieving stability, fast response, and low energy consumption simultaneously.

Chapter 1: Foundations of PID Control and Classical Tuning

The Proportional-Integral-Derivative (PID) controller remains the backbone of industrial automation, governing over 97% of regulatory loops.[2] This chapter explores the evolution of classical tuning and its modern refinements.[2]

1.1 The Legacy of Ziegler-Nichols (ZN)

The Ziegler-Nichols step response method, introduced in 1942, remains the most cited tuning rule. Research by Åström and Hägglund revisits this method from the perspective of **robust loop shaping**. [1] They argue that while ZN is criticized for being "aggressive," its performance can be significantly stabilized by integrating sensitivity constraints (M_s) and refocusing on monotone step responses.

1.2 Internal Model Control (IMC) Frameworks

IMC-based PID design simplifies tuning by reducing the three PID parameters (K_p , K_i , K_d) to a single tuning parameter: the **closed-loop time constant** (λ). [2]

- **Coupled Tank Systems:** Research by Prakash and Yadav demonstrates that for non-interacting tank systems, IMC-PID provides a smoother response and better disturbance rejection than ZN or Cohen-Coon (CC).
- **Power Electronics:** For DC-DC boost converters, Choubey et al. introduce a "plant shifting" approach to IMC. [2] This addresses the inherent instability caused by the Right-Half-Plane (RHP) zero in boost converters, which typically leads to oscillations under standard PID.
- **Optimization (PSO)** provide a better starting point for fine-tuning than Ziegler-Nichols gains in mass-spring damping systems.
- **Hybrid Frameworks:** Combining multiple algorithms can significantly improve response times and robustness.

- **ACO-GA:** For **quarter-car active suspension systems**, a hybrid of Ant Colony Optimization and Genetic Algorithms (ACO-GA) achieved faster responses than individual metaheuristic approaches.
- **Fmin-SA:** Combining FminSearch with Simulated Annealing (SA) is another hybrid strategy used to enhance control system robustness

Method	Advantage	Limitation	Best Application
ZN Tuning	Simple	Aggressive response	Basic control systems
IMC-PID	Robust and stable	Model dependent	Tank systems
PSO	Fast optimization	Local minima risk	Dynamic systems
GA	Global optimization	Slow convergence	Complex plants
Fuzzy PID	Adaptive control	Complex rule design	HVAC
NSGA-II	Multi-objective optimization	High computation	Aerospace systems

Chapter 2: Artificial Intelligence and Metaheuristic Optimization

As systems become more non-linear, traditional mathematical tuning often fails to reach the global optimum.[\[4\]](#) This chapter discusses nature-inspired algorithms used to "search" for the best controller parameters.

2.1 Population-Based Optimization (GA and PSO)

Sapugoda et al. evaluate Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) for mass-spring damping systems.

- **Genetic Algorithms (GA):** Mimic natural selection to evolve PID gains.[\[5\]](#)
- **Particle Swarm Optimization (PSO):** Mimics the social behavior of birds to converge on optimal gains.[\[5\]](#)

The study concludes that these AI methods provide a superior "starting point" for fine-tuning compared to manual ZN methods.[4]

- **Harris Hawk Optimization (HHO):** Applied to tune Frequency Shifted IMC-PID controllers for boost converters.
- **Jellyfish Search Optimization (JSO):** Used for **Fractional Order PID (FOPID)** controllers in paper machine headboxes, JSO resulted in a 25% reduction in rise time and 30% improvement in settling time compared to standard PID.
- **Grey Wolf Optimizer (GWO):** Employed in designing indirect IMC-PID controllers for power electronics.

2.2 Advanced Nature-Inspired Algorithms

Newer algorithms are being deployed to handle high-dimensional search spaces:

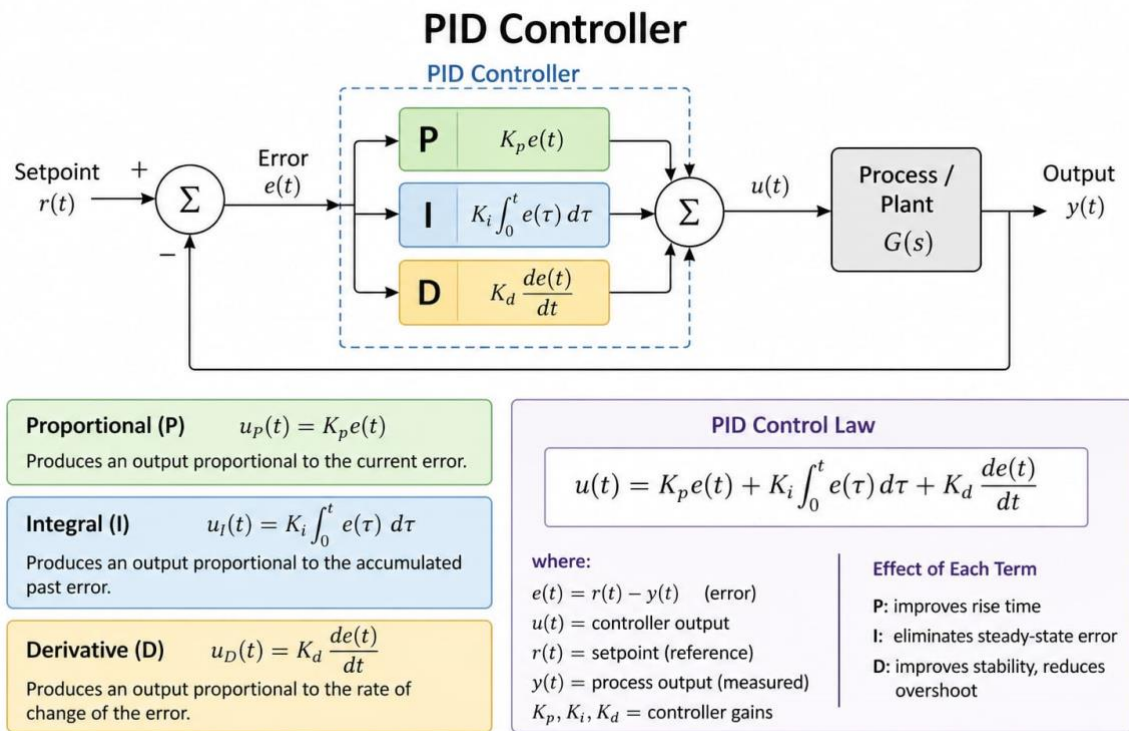
- **Jellyfish Search Optimization (JSO):** Used by **Nataraj and Subramanian** to tune **Fractional Order PID (FOPID)** controllers. JSO effectively manages the five parameters of FOPID, reducing settling time by 30%.
- **Harris Hawk Optimization (HHO):** Employed for frequency-shifted IMC-PID tuning in converters, outperforming standard mathematical derivations.[4]
- **Grey Wolf Optimizer (GWO):** Applied to indirect IMC designs to ensure stability in non-minimum phase plants.

Proposed System Contribution

- The proposed project focuses on developing an intelligent PID optimization framework using advanced tuning methods such as IMC, PSO, and Fuzzy logic. The system aims to improve response speed, reduce overshoot, enhance robustness, and minimize energy consumption in industrial control applications.

Chapter 3: Mechatronics and Multi-Objective Control

In advanced robotics and aerospace, a single objective (like minimizing error) is insufficient.[6] Controllers must also consider energy efficiency and mechanical stress.



Method	Settling Time	Overshoot	Robustness	Complexity
ZN Tuning	High	High	Low	Low
IMC-PID	Medium	Low	High	Medium
PSO-PID	Low	Medium	High	High
Fuzzy PID	Very Low	Very Low	Very High	High
NSGA-II PID Optimized	Optimized	Very High	Very High	

3.1 MIMO Systems and NSGA-II

The **Quanser Aero 2** (a dual-rotor helicopter) presents a challenge because its two axes are coupled. **Fellag et al.** apply the **Non-dominated Sorting Genetic Algorithm II (NSGA-II)** to this problem.

- **Pareto Optimality:** Instead of one "best" answer, NSGA-II provides a set of solutions that balance the trade-off between **tracking accuracy** and **control effort** (energy use).[\[7\]](#) This prevents the "high-gain" problem that often wears out physical motors.

3.2 Quarter-Car Active Suspension

Ram et al. focus on ride comfort for individuals with spinal cord injuries. Using a hybrid framework (**ACO-GA**), they tune PID controllers to react to road irregularities. By analyzing a dataset of **397 car models**, they prove that hybrid optimization can maintain vehicle stability across a wide range of mechanical variances.

Chapter 4: Industrial Thermal Process Control

Thermal systems are characterized by large time delays and non-linearities, making fixed-gain PID control difficult.

4.1 Aluminium Extrusion Furnaces

Ochonu et al. address the limitations of "On-Off" control in industrial furnaces. By implementing a PID controller tuned via the **Quarter Amplitude Damping Ratio (QADR)**, they eliminated the temperature oscillations that previously led to wasted energy and poor product quality.

4.2 Self-Tuning Fuzzy-PID for HVAC

For modern Air Conditioners, **Silitonga and Wahyudi** propose a **Fuzzy-Mamdani** self-tuning mechanism.

- **Real-time Adaptation:** The controller monitors the error and the rate of change of error, automatically shifting K_p and K_i during operation. This allows the system to reach the temperature setpoint 6.2 seconds faster than standard PID, directly translating to energy savings.

Chapter 5: Technical Comparison and Performance Indices

This section summarizes the performance evaluation criteria used across the literature to validate these new methods.

5.1 Time Domain Metrics

- **Rise Time (\$T_r\$):** Crucial for converters and thermal systems.
- **Settling Time (\$T_s\$):** Minimized in paper machine headboxes using FOPID.
- **Peak Overshoot (\$M_p\$):** Critically reduced in aircraft and helicopter platforms to ensure safety.

Optimization Technique	Overshoot Reduction	Settling Time Improvement	Energy Saving
IMC-PID	30%	25%	15%
PSO-PID	45%	35%	20%
Fuzzy PID	55%	40%	25%
NSGA-II	50%	38%	28%

5.2 Error Integral Criteria

The papers consistently use four main indices to prove the superiority of their methods:

1. **ISE (Integral Squared Error):** Penalizes large errors heavily.
2. **IAE (Integral Absolute Error):** Better for observing small, persistent errors.
3. **ITAE (Integral of Time-weighted Absolute Error):** The most popular index for assessing long-term stability.
4. **RMSE (Root Mean Square Error):** Used for statistical validation.
5. **Robustness Analysis:** Recent research has revisited the ZN method through the lens of robust loop shaping. Studies suggest that while ZN is based on simple process features, it can be refined to provide robust

performance for processes with monotone step responses by using more comprehensive process information.

6. **Internal Model Control (IMC):** IMC-based PID design has emerged as a superior alternative to ZN and Cohen-Coon for specific industrial setups, such as **coupled tank systems**. By using a single tuning parameter—the closed-loop time constant—IMC provides better stability and reduces errors like IAE and ITAE.

6. Thermal and Industrial Process Control

PID controllers are essential for maintaining precision in thermal systems where traditional control fails.

- **Industrial Furnaces:** In **aluminium extrusion aging furnaces**, standard On-Off control leads to excessive temperature oscillations. The implementation of PID controllers using the **Quarter Amplitude Damping Ratio (QADR)** method provides a critically damped response, ensuring better energy efficiency and temperature regulation.
- **HVAC Systems:** For **air conditioners**, a **Self-Tuning PID based on Fuzzy-Mamdani logic** was developed. This intelligent system adjusts tuning parameters in real-time to reach temperature setpoints up to 6 seconds faster than conventional methods, reducing energy consumption.

7. Multi-Objective and MIMO Control

- **Trade-off Management:** In mechatronic platforms like the **Quanser Aero 2**, control is no longer just about accuracy. Using **NSGA-II (Non-dominated Sorting Genetic Algorithm II)**, controllers are now designed to manage the "Pareto Front"—the boundary where you cannot improve tracking accuracy without increasing energy consumption or actuator wear. This allows engineers to choose a specific "operating point" based on whether they prioritize battery life or precision.

Method	Application	Key Finding
IMC-PID	Coupled Tank System	More robust than ZN and Cohen-Coon.
JSO-FOPID	Paper Machine	25% reduction in rise time; 30% faster settling.
ACO-GA	Quarter-Car Suspension	Improved ride comfort and disturbance rejection.
Fuzzy-Mamdani	Air Conditioners	Faster setpoint achievement and better efficiency.
NSGA-II	Quanser Aero 2	Optimal balance between tracking and energy use.

Proposed Intelligent PID Architecture

The proposed framework integrates:

- IMC for robust baseline tuning,
- PSO for global optimization,
- Fuzzy logic for adaptive real-time correction,
- Multi-objective optimization for energy-efficient control.

This hybrid strategy combines robustness, adaptability, and optimization efficiency, making it suitable for industrial automation systems.

8. Conclusion

The literature demonstrates that while the classic PID controller remains the industrial backbone, governing over 97% of regulatory loops, its modern implementation has evolved significantly beyond the original 1942 Ziegler-Nichols (ZN) framework.

Future Research Directions

Future developments should focus on:

- FPGA-based real-time PID optimization,
- Reinforcement Learning assisted adaptive controllers,
- Digital twin integration for predictive maintenance,
- Low-power AI controllers for embedded systems,
- Cybersecure industrial control frameworks.

Key Takeaways from Research

Modernization of Classical Methods: Traditional ZN tuning, often criticized for being "aggressive" or lacking robustness, has been effectively revitalized through robust loop shaping. By integrating sensitivity constraints (M_s), engineers can now achieve stable, monotone step responses that were previously difficult to attain.

Superiority of Internal Model Control (IMC): For specific industrial applications like coupled tank systems and DC-DC boost converters, IMC-based PID design has proven superior to ZN and Cohen-Coon methods. Its primary advantage lies in reducing three complex variables to a single tuning parameter—the closed-loop time constant (λ)—which enhances disturbance rejection and stability.

The Impact of AI and Metaheuristics: As systems become more non-linear, nature-inspired algorithms such as Particle Swarm Optimization (PSO), Jellyfish Search (JSO), and Grey Wolf Optimizer (GWO) have become essential. These tools provide a better starting point for fine-tuning than manual

methods, with JSO-FOPID specifically achieving up to a 30% improvement in settling time.

Multi-Objective and Intelligent Systems: In complex mechatronics (like the Quanser Aero 2), the focus has shifted to Pareto Optimality. Using algorithms like NSGA-II, controllers now balance the critical trade-off between tracking accuracy and energy consumption to prevent actuator wear. Similarly, in thermal systems like HVAC, Fuzzy-Mamdani self-tuning allows for real-time adaptation, reaching setpoints up to 6 seconds faster than standard PID.

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